Type MS · Adding up a finite geometric series

Answer 1 of 3 · Question 7 part c · 4 marks

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7 (a)	(i) $\frac{U_7}{U_3}=\frac{ar^6}{ar^2}=r^4\Rightarrow\frac{16}{4}=r^4\Rightarrow 4=r^4\Rightarrow 2=r^2\Rightarrow r=\sqrt{2}*$ (ii) $ar^2=4\Rightarrow a=\frac{4}{\left(\sqrt{2}\right)^2}=2$	M1 A1cso A1 [3]
Alternative		
(a)	(i) $ar^2=4\ ,\ ar^6=16\Rightarrow a\left(\frac{4}{a}\right)^3=16\Rightarrow a^2=4\Rightarrow a=2$ $r^2=\frac{4}{2}\Rightarrow r=\sqrt{2}*$ (ii) $a=2$	M1 A1cso A1
(b)	$500<2\times\sqrt{2}^{(n-1)}\Rightarrow 250<\sqrt{2}^{(n-1)}$ or $500<2^{\frac{1}{2}(n+1)}$ $\lg 250<(n-1)\lg\sqrt{2}$ or $\lg 500<\frac{1}{2}(n+1)\lg 2$ $n-1>\frac{\lg 250}{\lg\sqrt{2}}\Rightarrow n-1>15.93...$ or $\frac{1}{2}(n+1)>\frac{\lg 500}{\lg 2}\Rightarrow\frac{1}{2}(n+1)>8.96...$ $n>16.93...\Rightarrow n=17$	M1 M1 M1A1 [4]
(c)	$S_{20}=\frac{2\left(\left(\sqrt{2}\right)^{20}-1\right)}{\sqrt{2}-1}=\frac{2046}{\sqrt{2}-1}$ $\frac{2046}{\left(\sqrt{2}-1\right)}\times\frac{\left(\sqrt{2}+1\right)}{\left(\sqrt{2}+1\right)}=2046\sqrt{2}+2046=2046\left(1+\sqrt{2}\right)$	M1A1 M1A1 [4]
		Total 11 marks

(a)(i)	M1	$ar^2=4$ and $ar^6=16$
	A1 cso	Correct working to eliminate a and show that $r=\sqrt{2}$ (Answer given) Must see $r^4=4$.
(ii)	A1	$a=2$ (NB: This is not a B mark; must have scored the first M mark.)
Alternative		
(a)(i)	M1	$ar^2=4$ and $ar^6=16$
	A1 cso	Correct working to find a and use it to show that $r=\sqrt{2}$ (Answer given) Must see a correct substitution to eliminate r which is processed correctly to give $a=2$, and $r^2=\frac{4}{2}$ or $r^6=\frac{16}{2}$.
(ii)	A1	$a=2$ (Must have scored the first M mark.)

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(b)	M1	$500 < 2\times\left(\sqrt{2}\right)^{(n-1)}$ Accept an equation or >. Accept $500 < 2\times\sqrt{2}^{n-1}$ ft their value for a
	M1	$n-1 > \frac{\lg 250}{\lg\sqrt{2}}$ or $\frac{1}{2}(n-1) > \frac{\lg 250}{\lg 2}$ or $\frac{1}{2}(n+1) > \frac{\lg 500}{\lg 2}$ or $\lg_{\sqrt{2}} 250 < n-1$ Accept equation or >, and logs with any base. ft $500 < 2\times\left(\sqrt{2}\right)^{n}$
	M1	Evaluate and divide logs and attempt to find a value (at least 1DP) or numerical expression for n. Ignore incorrect inequality sign. eg $n > 16.9$ or $n = 16.9$ or $n > 1+15.93$ or $n = 1+2\times 7.96$ ft $500 < 2\times\left(\sqrt{2}\right)^{n}$
	A1	$n = 17$ Do not award after incorrect inequalities e.g. $n < 16.9$
Alternative		
(b)	M1	$500 < 2\times\left(\sqrt{2}\right)^{(n-1)}$ Accept an equation or >. Accept $500 < 2\times\sqrt{2}^{n-1}$ ft their value for a
	M1	Attempt to use trial and improvement, showing two trials that can be used to confirm the required value of n, eg $\left(\sqrt{2}\right)^{15} = 181.0...$ and $\left(\sqrt{2}\right)^{16} = 256$ or $2\left(\sqrt{2}\right)^{15} = 362.0...$ and $2\left(\sqrt{2}\right)^{16} = 512$ or $2^8 = 256$ and $2^9 = 512$ ft $500 < 2\times\left(\sqrt{2}\right)^{n}$ if clearly stated
	M1	Use the higher power from these trials in an appropriate equation eg $n-1 = 16$ or $\frac{1}{2}(n+1) = 9$ and attempt to find a value for n. ft $500 < 2\times\left(\sqrt{2}\right)^{n}$ if clearly stated
	A1	$n = 17$
(c)	M1	Attempt to use $S_n = \frac{a(1-r^n)}{(1-r)}$ with their a, $r = \sqrt{2}$ and $n = 20$ Allow notation or sign errors if formula is stated, otherwise the substitution must be correct.
	A1	$S_{20} = \frac{2046}{\sqrt{2}-1}$ or $S_{20} = -\frac{2046}{1-\sqrt{2}}$ Correct expression with numerator evaluated.
	M1	Attempt to rationalise the denominator. Must use $\sqrt{2}+1$ and attempt to simplify the denominator.
	A1	$2046(1+\sqrt{2})$ or $2046(\sqrt{2}+1)$ Accept $p = 2046$. Award full marks for a correct answer if first M1 has been awarded.

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Question number	Scheme	Marks

Type MS · Adding up a finite geometric series

Answer 2 of 3 · Question 4 part i · 4 marks

2025 JUN · Paper 02 (regional R) · extract 1 of 3 · find part i in this mark scheme extract

Number		
4 (i)	$a=3$ or $r=2$	B1
	$\left(\sum_{k=1}^{17}3(2)^{k-1}\right)=\frac{3\left(1-(2)^{17}\right)}{1-2}=(393213)$ $\left(\sum_{k=1}^{5}3(2)^{k-1}\right)=\frac{3\left(1-(2)^{5}\right)}{1-2}=(93)$	M1
	"393213"–"93"	dM1
	393120	A1
		[4]
ALT	$a=3\times2^5\,(=96), r=2$	B1
	$\left(\sum_{k=6}^{17}3(2)^{k-1}\right)=\frac{\left(3\times2^5\right)\left(1-(2)^{\text{"11 or 12"}}\right)}{1-2}$	M1
	$\frac{\left(3\times2^5\right)\left(1-(2)^{12}\right)}{1-2}$	M1
	393120	A1
		[4]
(ii)	$ar^2=\frac{7}{13}$ or $ar^8=\frac{448}{9477}$	M1
	$\frac{ar^8}{ar^2}=\frac{\frac{448}{9477}}{\frac{7}{13}}\left(=\frac{64}{729}\right)\Rightarrow r^6=$	dM1
	$r=\sqrt[6]{\left(\frac{\frac{448}{9477}}{\frac{7}{13}}\right)}$	M1
	$r=\frac{2}{3}$ $a=\frac{63}{52}$	A1 A1
	$S_\infty=\frac{\text{"}\frac{63}{52}\text{"}}{1-\text{"}\frac{2}{3}\text{"}}$	M1
	$=\frac{189}{52}$ oe	A1
		[7]
		Total 11 marks

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Answer 2 of 3 · Question 4 part i · 4 marks

2025 JUN · Paper 02 (regional R) · extract 2 of 3 · find part i in this mark scheme extract

Part	Mark	Additional Guidance
(i)	B1	For either value of a or r correct, stated or implicit in working.
	M1	For correctly substituting into the formula for either sum. Allow use of their a and their r with any numbers it is clear they intend to be a and r
	dM1	For their 2 sums subtracted, the correct way round. Dependent on the previous method mark. **They must be using $a = 3$ for this mark.**
	A1	For 393120
ALT	B1	For either value of a or r correct, stated or implicit in working.
	M2	For correctly substituting into the formula for the sum to n terms. **Must be using $n = 12$ and $a = 96$, may use their r.** For M1 students may use $n = 11$ or 12 and allow use of their a and their r with any numbers it is clear they intend to be a and r
	A1	For 393120
	Students who complete this by adding every term B1 for the first term correct M2 for all terms written out and added (we are not expected to evaluate them), there is no M1 mark A1 correct answer and as usual, can imply the M and B marks. **A fully correct answer with little or no working will score 4 marks**	
(ii)	M1	For either correct equation.
	dM1	For correctly using their equations to eliminate a (or r) We will condone $\frac{ar^9}{ar^3}\left(=\frac{\frac{448}{9477}}{\frac{7}{13}}=\frac{64}{729}\right)\Rightarrow r^6=$ For this mark if the candidate works correctly to find r
	M1	Rearranges their equation, allowing one processing error, to find a value for r (or a). Do not award the mark if a student moves from r^6 to r without showing what they are doing if r is incorrect. We will condone correct r from $\frac{ar^9}{ar^3}\left(=\frac{\frac{448}{9477}}{\frac{7}{13}}=\frac{64}{729}\right)\Rightarrow r=$ if the candidate works correctly to find r
	A1	Either r or a correct.
	A1	Both r and a correct.
	M1	Correct substitution of their values for a and r into a correct formula for S_∞
	A1	$\frac{189}{52}$ oe

Type MS · Adding up a finite geometric series

Answer 2 of 3 · Question 4 part i · 4 marks

2025 JUN · Paper 02 (regional R) · extract 3 of 3 · find part i in this mark scheme extract

Question number	Scheme	Marks

Type MS · Adding up a finite geometric series

Answer 3 of 3 · Question 8 part b · 3 marks

2023 JUN · Paper 01 (regional R) · extract 1 of 3 · find part b in this mark scheme extract

8(a)	$U_5 = \left(\frac{25}{4}\right)\left(\frac{3}{5}\right)^5 = \frac{243}{500}$ oe	B1 [1]
(b)	$U_1 = \left(\frac{25}{4}\right)\left(\frac{3}{5}\right)^1 = \frac{15}{4}$ oe	B1
	$S_n = \left(\frac{15}{4}\right)\left(\frac{3}{5}\right)^0 + \left(\frac{15}{4}\right)\left(\frac{3}{5}\right)^1 + \ldots\ldots + \left(\frac{15}{4}\right)\left(\frac{3}{5}\right)^{n-1}$	M1
	$\Rightarrow S_n = \sum_{r=1}^{n}\left(\frac{15}{4}\right)\left(\frac{3}{5}\right)^{r-1}$ oe	A1 [3]
	ALT 1	
	$U_1 = \left(\frac{25}{4}\right)\left(\frac{3}{5}\right)^1 = \frac{15}{4}$	[B1
	$r = \frac{3}{5} \qquad U_n = \frac{15}{4} \times \left(\frac{3}{5}\right)^{n-1}$	M1
	$S_n = \sum_{r=1}^{n} \frac{15}{4} \times \left(\frac{3}{5}\right)^{r-1}$	A1]
	ALT 2	
	$S_n = \sum_{r=1}^{n} \frac{25}{4}\left(\frac{3}{5}\right)^r$	[B1
	$= \sum_{r=1}^{n} \frac{25}{4} \times \frac{3}{5} \times \left(\frac{3}{5}\right)^{r-1}$	M1
	$= \sum_{r=1}^{n} \frac{15}{4} \times \left(\frac{3}{5}\right)^{r-1}$	A1]
(c)	$S_\infty = \frac{\frac{15}{4}}{1-\frac{3}{5}} = \frac{75}{8}$ or $9.375 \qquad S_n = \frac{\frac{15}{4}\left(1-\left[\frac{3}{5}\right]^n\right)}{1-\frac{3}{5}}$	M1M1
	$9.375 - \frac{\frac{15}{4}\left(1-\left[\frac{3}{5}\right]^n\right)}{1-\frac{3}{5}} < 0.045 \Rightarrow \frac{\frac{15}{4}\left(1-\left[\frac{3}{5}\right]^n\right)}{1-\frac{3}{5}} > 9.33$	
	$\frac{\frac{15}{4}\left(1-\left[\frac{3}{5}\right]^n\right)}{1-\frac{3}{5}} > 9.33 \Rightarrow 1-\left[\frac{3}{5}\right]^n > 0.9952 \Rightarrow \left[\frac{3}{5}\right]^n < \frac{3}{625}$	dM1
	$n \lg\left[\frac{3}{5}\right] < \lg\frac{3}{625} \Rightarrow n > 10.45$	ddM1A1
	$n = 11$	A1 [6]
		Total 10 marks

Type MS · Adding up a finite geometric series

Answer 3 of 3 · Question 8 part b · 3 marks

2023 JUN · Paper 01 (regional R) · extract 2 of 3 · find part b in this mark scheme extract

Part	Mark	Notes
		Mark (a) and (b) together
(a)	B1	For the correct value only. Decimal value is 0.486.
(b)	B1	Finds the value of the first term. Must be identified as U_1 / first term.
	M1	Forms the sum of the series with at least the first term and the nth term.
	A1	For the correct expression.
	ALT 1	
	B1	Finds the value of the first term. Must be identified as U_1 / first term.
	M1	For finding r and writing $U_n = \frac{15}{4} \times \left(\frac{3}{5}\right)^{n-1}$
	A1	For the correct expression.
	ALT 2	
	B1	For correct summation statement using U_n from the question
	M1	For showing process to change index from r to $r-1$ and find the value of $\frac{A}{B}$ Minimum working is $\frac{25}{4} \times \left(\frac{3}{5}\right)^{r} = \frac{25}{4} \times \frac{3}{5} \times \left(\frac{3}{5}\right)^{r-1}$. May be with n as index.
	A1	For the correct expression.
(c)	Trial and improvement approaches for (c) should be sent to review.	
	M1	Correct method to find the sum to infinity using the correct formula with their $a = \frac{15}{4}$ and $r = \frac{3}{5}$. Their a should be $\frac{A}{B}$ if found in (b). Condone if working with $a = \frac{25}{4}$, r must be correct.
	M1	Using the correct formula, forms the sum to n terms using their $a = \frac{15}{4}$ and $r = \frac{3}{5}$. Their a should be $\frac{A}{B}$ if found in (b). Condone if working with $a = \frac{25}{4}$, r must be correct.
	dM1	Forms the inequality in terms of n with their S_∞ and S_n and simplifies to $\left[\frac{3}{5}\right]^n < '\frac{3}{625}'$ Must have dealt with negative. Allow for working with equation if inequality sign reinstated later. Reinstatement can be implied by rounding up to the next integer value once n found. Dep M1M1
	ddM1	Uses logarithms correctly to attempt to find the value of n They must reverse the inequality for this mark. Allow for working with equation if inequality sign reinstated later. Reinstatement can be implied by rounding up to the next integer value once n found. Dep previous M mark. Allow use of log base $\frac{3}{5}$.
	A1	For correct n awrt 10.5 Can be implied by $n = 11$ if decimal not seen.
	A1	For $n = 11$

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Answer 3 of 3 · Question 8 part b · 3 marks

2023 JUN · Paper 01 (regional R) · extract 3 of 3 · find part b in this mark scheme extract

Question	Scheme	Marks